

EXPERIMENT – 02(a)

Experiment No.	02(a)
Date	_____
Title	Java Program to Check Whether the Given Number is Prime Palindrome or Not

1. Objective / Aim

To write a Java program to check whether a given number is both prime and palindrome.

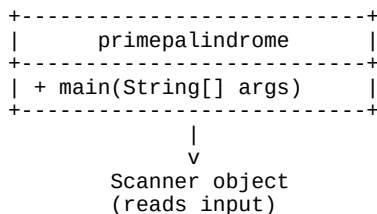
2. System Requirements

Hardware	Computer/Laptop with keyboard and monitor
Software	JDK (Java Development Kit), any Java IDE or Command Prompt

3. Algorithm / Logic Description

1. Read an integer n from the user.
2. Check whether n is prime by testing divisibility from 2 to $n - 2$.
3. Reverse the digits of n using the remainder (%) and division (/) operators.
4. Compare the original number with its reversed number.
5. If the number is prime and the original number equals its reverse, display that it is a prime palindrome.
6. Otherwise, display that it is not a prime palindrome.

4. UML Class Diagram



5. Source Code

```
import java.util.Scanner;

public class primepalindrome {
    public static void main(String[] args) {
        int n, i, prime = 0;

        Scanner sc = new Scanner(System.in);

        System.out.println("enter a number");
        n = sc.nextInt();

        if (n < 2) {
            prime = 1;
        }

        for (i = 2; i < n - 1; i++) {
            if (n % i == 0) {
                prime = 1;
                break;
            }
        }

        int temp = n;
        int rev = 0;

        while (temp > 0) {
            int dig = temp % 10;
```

```

        rev = rev * 10 + dig;
        temp = temp / 10;
    }

    if (prime == 0 && n == rev) {
        System.out.println("GIVEN NUMBER IS PRIME PALINDROME");
    } else {
        System.out.println("given number is not prime palindrome number");
    }
}
}

```

6. Compilation & Execution Commands

```

javac primepalindrome.java
java primepalindrome

```

7. Output Screen

Test Case 1 – Prime Palindrome

```

enter a number
131
GIVEN NUMBER IS PRIME PALINDROME

```

Test Case 2 – Not a Prime Palindrome

```

enter a number
121
given number is not prime palindrome number

```

Code Check

The logic in the submitted code is essentially correct for normal positive inputs. The version above keeps the same structure and class name, while adding an explicit check for numbers below 2 because 0 and 1 are not prime. The prime test can also be optimized later using $i \times i \leq n$, but that is not required for this record.